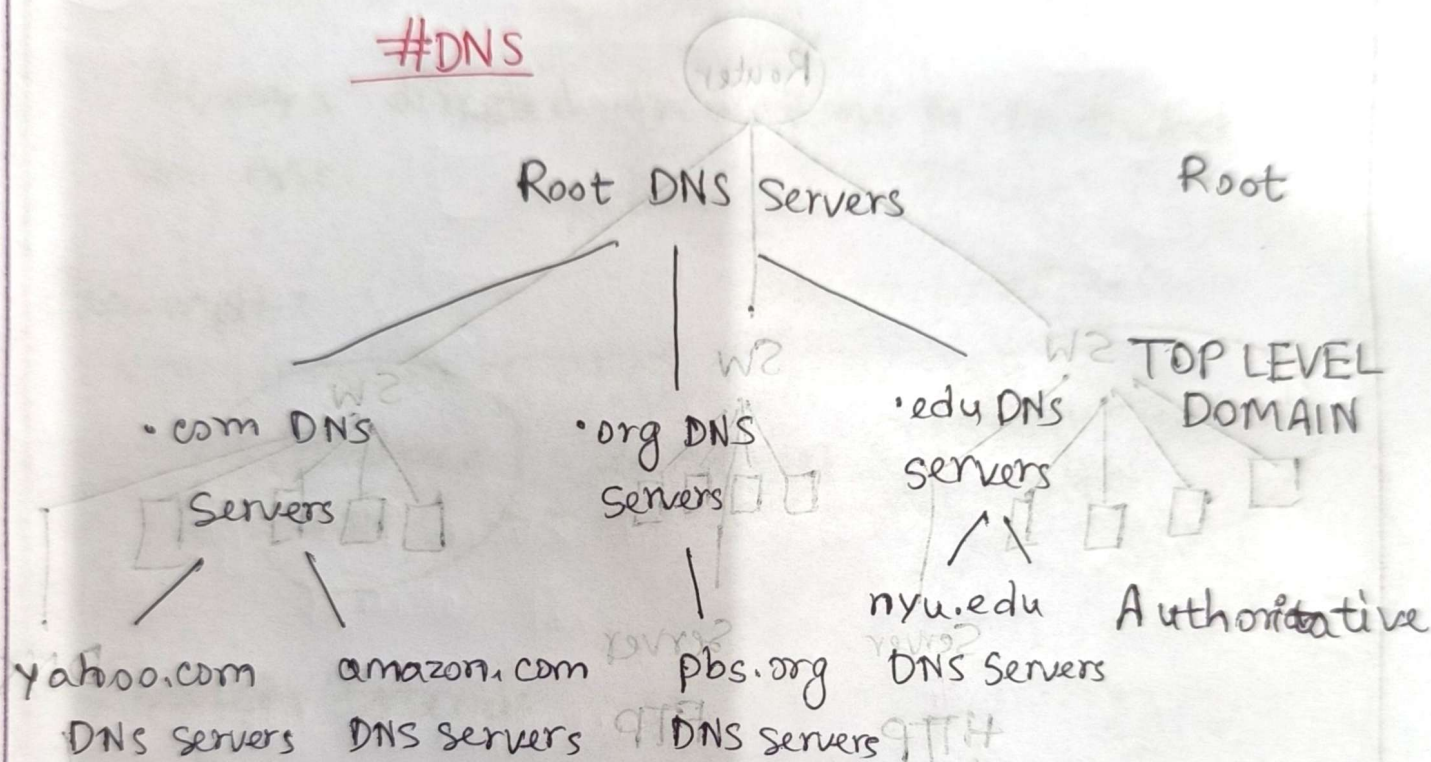


Computer Networks | class - 07

TCP
UDP
IP Address
Cryptography

প্রায় অধিক পরিমাণে

#DNS



Client wants IP Address for `www.amazon.com`;
1st approximation:

☐ client queries root server to find `.com` DNS server

☐ > {from slide}

{ DNS is managed in tree structure }

Top Level Domain

Definition

Authoritative DNS

Definition

Queries

Iterative
(Iterated)

Recursive

Real world a mixed use
ਕਰਾ 221

ns lookup → find IP from hostname

ping

→ using either iterated or recursive

DNS Records

Book: Page 140

Resource Record:

RR format: (name, value, type, ttl)

'Type=A' record = Address Record

'Type=NS' record = Name-Server

Time to Live
of resource record

'Type=CNAME' record = Canonical / Alias Hostname Record

'Type=MX' record = Canonical Name of a Mail Server
that has an alias hostname
'Name'

Final Assignment

Code for building an HTTP/FTP server

FTP (easy)

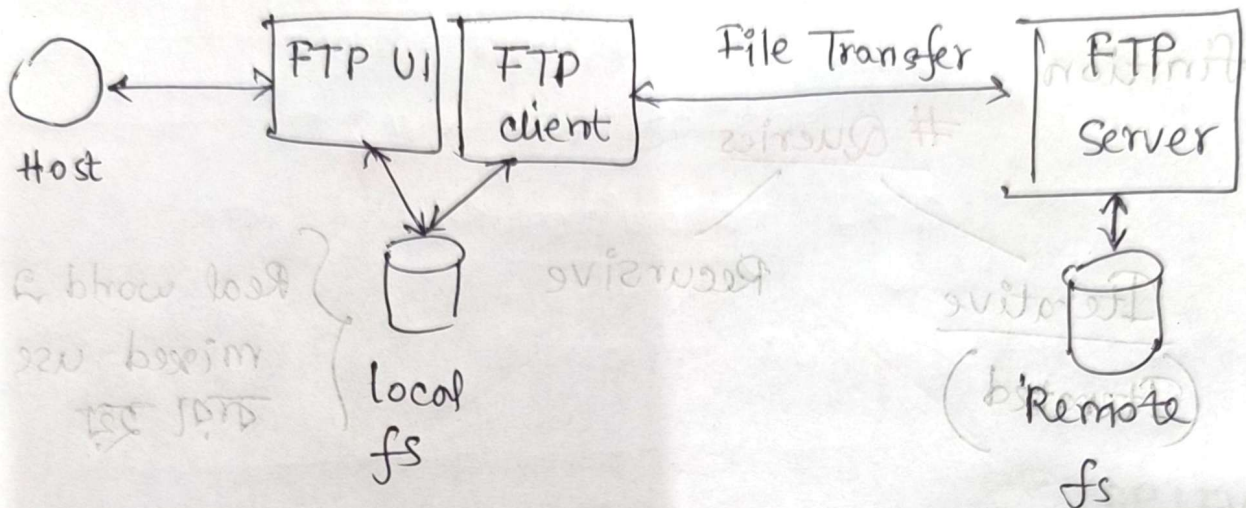


fig: FTP moves files between local and remote fs.

File Transfer line always kept alive

↳ port 21 → control connection

↳ port 20 → data connection

'Inband'

'Out-of-Band'

Common FTP commands

↓
SMTP/HTTP

↓
FTP

- ① USER ② PASS
- ③ LIST(dir)
- ④ RETR(get)
- ⑤ STOR(put)

Status codes

- 331: Username OK, password required
- 125: Data connection already open, transfer starting
- 425: Can't open data connection
- 452: Error writing file

#POP-IMAP

QAMI#

#POP3

↳ extremely simple mail access protocol

telnet mailServer 110

mail fetcher program
↳ ~~use~~ to download all mails
from Gmail

↳ look up online

↳ Do using either POP3 or IMAP

POP3 has 1 directory: "Inbox"

IMAP can contain many directories: "Inbox",
"Updates",
"Social",
etc.

list

1 498

2 912

<content>

dele 1 ← won't delete immediately

retr 2

<content>

dele 2

quit ← deletes before quitting

OK POP3 server signing off

IMAP

↳ can be read in partitions

↳ only header

↳ only specific part

↳ excluding LFD

↓
large file download

Nodal Delay / TCP

(নিজের কাজ নিয়ে)

IP Addresses

↳ 32 bits (IPv4)

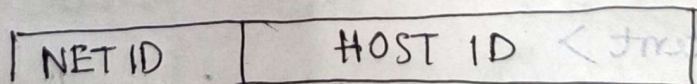
↳ 128 bits (IPv6)

2^{24} computers

3 byte = 24 bits

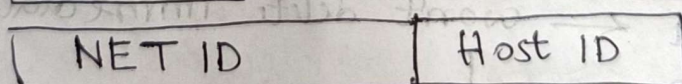
IPv4

Class A



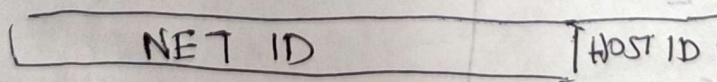
2^3

Class B



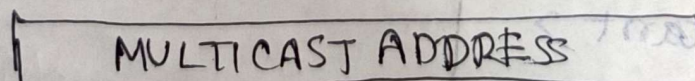
2^4

Class C



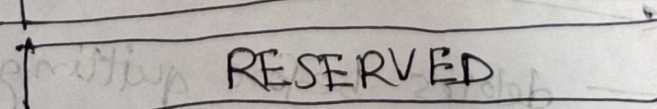
2^8

Class D

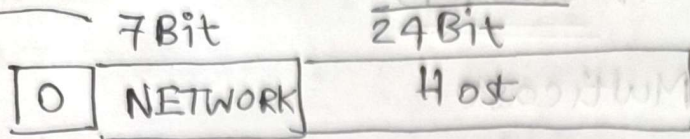


2^{16}

Class E



Class A



class A's IP Address first bit 0

$(2^{24} - 2)$ valid hosts

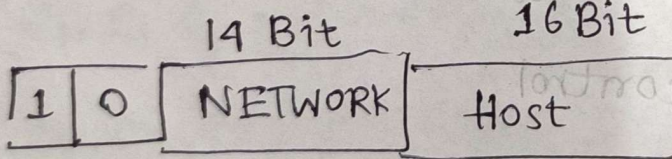
excluding $(0.0.0.0)$
 $(1.1.1.1)$

127.0.0.1

look back address

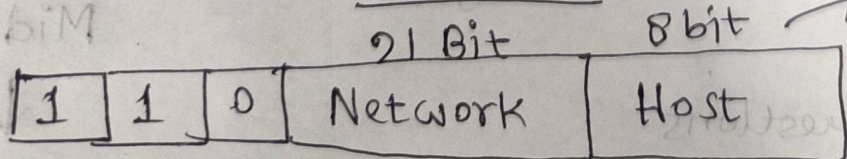
$(2^7 - 2)$ valid addresses

Class B



Class B

Class C



class C

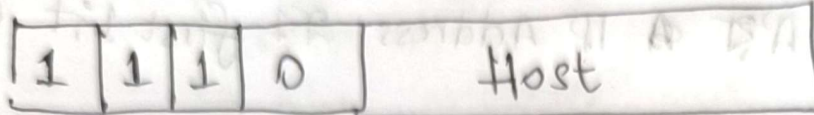
$(2^8 - 2)$ hosts

scale no. of networks and hosts

class D

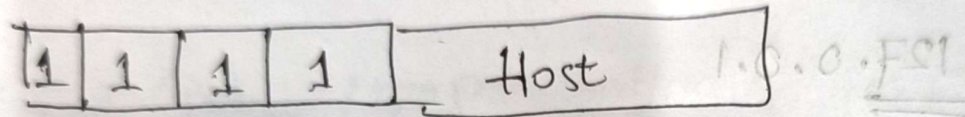
Multicast

28 bit



class E

Research purposes



Identify class by converting IP to binary

192.168.20.10 → class C

class A-C public use

Lab → access control

↳ IP address

↳ Subnetting

CT

↳ 2/3 Questions

↳ { 10 Marks }

Mid এর আগ

প্রতিফলন ($K_A = K_B$)

Cryptographic Components

Plaintext : a-z

Cipher text : shift 11 12

"A monoalphabetic cipher"

apple → mlqgc

shift 12 ~~indexes~~
indices/letters

"Polyalphabetic Cipher"

Plain Text : a-z

$C_1 (k=5)$ } Two ~~as~~ Caesar cipher
 $C_2 (k=19)$

$\langle C_1, C_1, C_1, C_2 \rangle$ sequence

$\langle C_1, C_2, C_1, C_2 \rangle$ sequence

$\langle C_1, C_2, C_1, C_2, C_1, C_2 \rangle$

apple	f	i	u	e	j
	C_1	C_2	C_1	C_2	C_1

{ "Mono and poly alphabetic cipher" }
"Caesar cipher" → examples ✓